Maharashtra State Board of Technical Education, Mumbai

A project Report on

Web-Based Hardware Marketplace for S. K. Infocom

Submitted in partial fulfillment for the award of

Diploma in Computer Engineering

By

Amey Pravin Dere

Shrutika Sandip Dhondge

Anushka Sangram Digole

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DEPARTMENT OF COMPUTER ENGINEERING

S. S. V. P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

(Institute Code : 0059)

2025-2026

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MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION, MUMBAI

S.S.V.P.S's Bapusaheb Shivajirao Deore Polytechnic, Dhule

DEPARTMET OF COMPUTER ENGINEERING

CERTIFICATE

This is to certify that,

Sr. No.	Roll No.	Name	Enrollment No.	Exam Seat No
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has satisfactorily completed project entitled “Web-Based Hardware Marketplace for S. K. Infocom”in partial fulfillment for the **Diploma Course in Computer Engineering** of the **Maharashtra State Board of Technical Education** at our polytechnic during the academic year **2025-2026**. The project is completed by a group consisting of four persons under the guidance of the Faculty Guide.

Date : **Place:** Dhule

Signature: **Signature:**

Name : **Name :**

Faculty Guide **External Examiner**

Signature: **Signature:**

Name : **Name :**

H.O.D. **Principal**.

Acknowledgement

Department for providing us with the opportunity and necessary support to complete our capstone project successfully. We are deeply thankful to our respected mentor, Mr. A.V. Agrawal sir, for her constant guidance, encouragement, and valuable suggestions throughout the project. Her support and motivation helped us overcome various challenges and improve the quality of our work.

We also extend our heartfelt thanks to our subject teacher, Mr. N. D. Patel Sir, for his important guidance, advice, and cooperation during the development of this project. We are equally grateful to all the faculty members of the Computer Department and our colleagues who directly or indirectly supported us with their ideas, suggestions, and encouragement in completing this project successfully.

Amey Pravin Dere,
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Sujal Nitin Pardeshi.

Abstract

The Web-Based Hardware Marketplace for S. K. Infocom is a web application designed to simplify and modernize the process of buying computer hardware products and accessing related technical services online. The main objective of this system is to provide a digital platform where customers can easily browse, compare, and purchase hardware components such as processors, motherboards, RAM, storage devices, and other computer accessories. In addition to product sales, the system also provides information about services related to computer hardware, such as repair, maintenance, installation, and technical support.

The application allows users to view product details, check availability, request services, and place orders through a simple and user-friendly interface. On the administrative side, the system enables the administrator to manage products, update service details, handle customer orders, and maintain inventory records efficiently. By implementing this web-based marketplace, S. K. Infocom can reduce manual work, improve business operations, and reach a wider range of customers. The system ensures organized data management, better accessibility, and a convenient platform for customers to purchase hardware products and request related services.

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CHAPTER 01

INTRODUCTION AND BACKGROUND OF INDUSTRY

1.1 INTRODUCTION :

In today's digital era, computer hardware plays a crucial role in almost every sector such as education, business, healthcare, and communication. With the rapid growth of technology, the demand for computer hardware components and related services has increased significantly. Customers require reliable platforms where they can easily purchase hardware products and access technical services. Traditional hardware stores mainly operate through offline methods, which can be time-consuming and less efficient for both customers and store owners.

The computer hardware industry includes the manufacturing, distribution, and sale of components such as processors, motherboards, RAM, storage devices, graphic cards, and various peripheral devices. In addition to selling products, many businesses in this industry also provide services such as computer repair, system maintenance, installation, and technical support. As technology continues to evolve, the industry is gradually shifting toward online platforms to improve accessibility, customer convenience, and business efficiency.

1.2 BACKGROUND OF INDUSTRY :

The computer hardware industry has grown rapidly with the increasing use of computers and digital technologies in various fields such as education, business, healthcare, and communication. Computer hardware includes the physical components of a computer system such as processors, motherboards, RAM, storage devices, graphic cards, and peripheral devices like keyboards, monitors, and printers. As technology continues to advance, the demand for high-performance hardware components and reliable technical services has increased significantly. Along with selling hardware products, many businesses in this industry also provide important services such as computer repair, hardware installation, system upgrades, troubleshooting, and maintenance. These services help customers maintain their computer systems and ensure smooth operation.

Traditionally, most hardware businesses operate through physical stores where customers must visit to purchase products or request services. However, this method has several limitations such as limited customer reach, manual record keeping, and difficulty in managing inventory and orders efficiently. With the growth of internet technology, many businesses are moving toward web-based platforms. A web-based hardware marketplace helps businesses digitalize their operations, allowing customers to browse products, request services, and place orders online while improving efficiency and overall customer experience.

- **Capstone Project :**

The Web-Based Hardware Marketplace for S. K. Infocom is an online platform designed to simplify the process of buying computer hardware products and accessing related services. Customers can easily browse available products, check their details and availability, place orders, and request services such as repair, installation, and maintenance. On the administrative side, the system allows the shop to manage products, track inventory, and handle customer orders and service requests efficiently. This project helps digitalize the traditional hardware store, providing a convenient experience for customers while improving overall business management and efficiency.

CHAPTER 02

LITERATURE SURVEY AND SPECIFICATION

2.1 STUDY OF EXISTING SYSTEM / REVIEW OF RESEARCH PAPER :

In the existing hardware retail and service systems, most businesses operate through traditional offline stores. Customers must physically visit the store to browse products, place orders, or request services. This process often involves manual record keeping for inventory, sales, and service requests, which can lead to errors, inefficiency, and delays. Customer reach is limited, and tracking orders or service status can be challenging.

Several research papers on e-commerce and online marketplaces highlight the advantages of web-based systems in improving business operations. Studies show that digital platforms allow real-time inventory management, easier order processing, and better customer engagement. Web-based marketplaces for hardware and related services help businesses expand their reach, reduce manual work, and provide convenient access for customers. These studies provide a foundation for developing a web-based hardware marketplace that integrates product sales with service requests, offering a more efficient and user-friendly system compared to traditional methods.

- **Limitations of Existing System ::**

  - Most hardware stores and management systems rely on manual record-keeping, leading to errors and inefficiency.
  - Customers often need to visit the store physically to purchase products or request services, reducing accessibility.
  - Existing e-commerce platforms like Magento or Shopify focus only on product sales, without integrated service management.
  - There is no centralized system to handle both hardware products and related technical services.

2.2 PROBLEM IDENTIFICATION / NEED OF A SYSTEM :

- **Problems Identified :**

1. Limited Customer Reach: Customers must visit the shop physically to see or purchase hardware products.
2. Manual Product Management: Product details, prices, and stock are managed manually which may cause errors.
3. Time-Consuming Order Process: Handling orders manually takes more time and may lead to delays.
4. Lack of Online Availability: Customers cannot check product availability or prices online.
5. Poor Inventory Tracking: It becomes difficult to track stock levels accurately.
6. Limited Marketing Opportunities: Without an online platform, it is harder to attract new customers.

- **Need for the System :**

1. Allow customers to view and purchase hardware products online.
2. Provide easy product management for the admin.
3. Enable real-time inventory tracking.
4. Make the ordering process faster and more efficient.
5. .Increase business reach and customer convenience

2.3 PROBLEM DEFINITION :

The current system of S. K. Infocom is mostly manual and requires customers to visit the shop to view and purchase hardware products and services. This makes the process time-consuming and inconvenient. It is also difficult for the shop owner to manage product details, repair services, orders, and stock manually. Due to the lack of an online platform, customers cannot check product availability or service options from home. Therefore, a web-based hardware marketplace is needed to manage products and services and allow customers to buy hardware or request services online easily.

CHAPTER 03

SCOPE OF PROJECT

3.1 SCOPE OF THE PROJECT :

The scope of the Web-Based Hardware Marketplace for S. K. Infocom is to develop an online platform that allows customers to easily view, compare, and purchase computer hardware products through a website. The system will provide detailed information about different hardware items such as computers, laptops, printers, accessories, and other electronic components. Customers will be able to browse products from anywhere without visiting the physical store, which improves convenience and saves time.

The system will also include features for managing various services related to hardware such as repair services, installation services, and product inquiries. Customers will be able to request these services through the website. This helps the business provide better support and maintain good communication with customers.

From the administration side, the system will help the shop owner manage products, update prices, track stock availability, and monitor customer orders efficiently. The admin can add new products, update existing product details, and remove unavailable items from the system. It will also help maintain a proper database of customers, orders, and services.

The web-based system will improve business operations by reducing manual work and improving data management. It will also help the business reach more customers through the internet and provide faster service. Overall, the system aims to make the process of buying hardware products and requesting services easier, faster, and more efficient for both customers and the business.

CHAPTER 04

METHODOLOGY

4.1 FUNCTIONAL REQUIREMENT :

There are three user roles for the system: Admin, Seller/Service Staff, and Customer.

i. Requirement of Admin:

- Admin shall be able to login with a unique username and password.
- Admin shall be able to add, update, and delete hardware products.
- Admin shall be able to manage product categories such as computers, accessories, and spare parts.
- Admin shall be able to view and manage customer orders.
- Admin shall be able to track product stock and inventory.
- Admin shall be able to manage hardware service requests such as repair or installation.
- Admin shall be able to view customer details and purchase records.
- Admin shall be able to generate reports of sales and services.

ii. Requirements of Seller / Service Staff:

- Seller shall be able to login with their username and password.
- Seller shall be able to update product details and prices.
- Seller shall be able to check customer orders assigned to them.
- Seller shall be able to update order status such as processing, shipped, or completed.
- Seller shall be able to view and respond to service requests from customers.
- Seller shall be able to manage repair or installation service updates.

iii. Requirements of Customer:

- Customer shall be able to register and login into the system.
- Customer shall be able to browse hardware products available on the website.
- Customer shall be able to view product details such as price, specification, and availability.
- Customer shall be able to add products to cart and place orders online.
- Customer shall be able to request services such as hardware repair or installation.

4.2 NON-FUNCTIONAL REQUIREMENTS :

- The system should be easy to use and user-friendly for customers and admin.
- The system should provide fast performance and quick loading of product pages.
- The system should ensure security of user accounts and customer data.
- The system should be reliable and work properly without frequent errors.
- The system should be available anytime for users to access the website.
- The system should support different web browsers like Chrome and Edge.
- The system should allow data backup and recovery to prevent data loss.
- The system should be scalable so that more products, services, and users can be added in the future.

4.3 ACTION PLAN :

- **Steps followed to developed project work monitoring system :**

- Finalization of project team and allocation of project guide
- Identification of project domain related to web-based hardware marketplace and online hardware services**.**
- Submission of project proposal for developing a Web-Based Hardware Marketplace for S. K. Infocom.
- Finalization of project idea by project guide and HOD.
- Conduct industrial survey of hardware shops and study existing online marketplaces.
- Literature survey on web technologies, hardware product management, and online service systems.
- Project identification and preparation of system specification.
- Proposed methodology for developing the web-based marketplace system.

- **Development Phase :**

- Preparation of project planning report and presentation.
- Presentation of project planning to guide and faculty.
- Submission of project planning report.
- Design of User Interface for product listing, order system, and service requests.

- Design database structure for products, customers, orders, and services and create database.
- Develop GUI and interfaces for admin panel and customer panel.
- Programming and coding of system modules using web technologies.
- Database connectivity for storing and retrieving product and order data.
- Installation of project and testing of system functions.
- Modifications and improvements based on suggestions given by project guide and teacher.
- Prepare final project presentation and compile draft copy of complete project report.
- Make corrections in draft copy of project report and presentation.
- Final submission of project report and term work

CHAPTER 05

DETAILS OF DESIGN WORKING AND PROCESSES

5.1 BLOCK DIAGRAM OF SYSTEM :

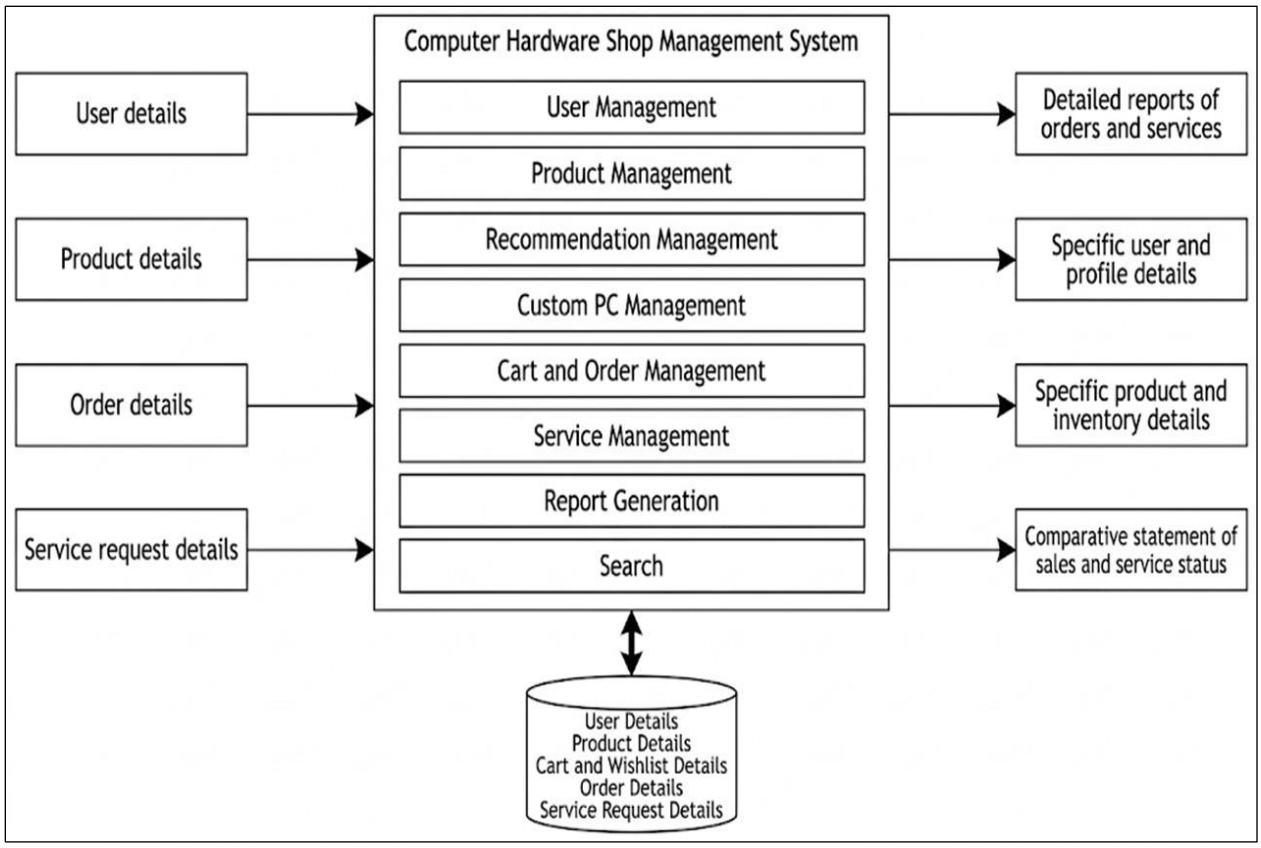


Fig 5.1. System Design

Fig 5.1 shows the diagram represents a Computer Hardware Shop Management System that helps manage the activities of a computer hardware shop in an organized way. The system receives input data such as user details, product details, order details, and service request details. It contains different modules like user management, product management, recommendation management, custom PC management, cart and order management, service management, report generation, and search. These modules help in managing users, products, orders, services, and product searches.All the information is stored in a central database that includes user details, product details, cart and Wishlist data, order details, and service requests.

5.2 USE CASE DIAGRAM :

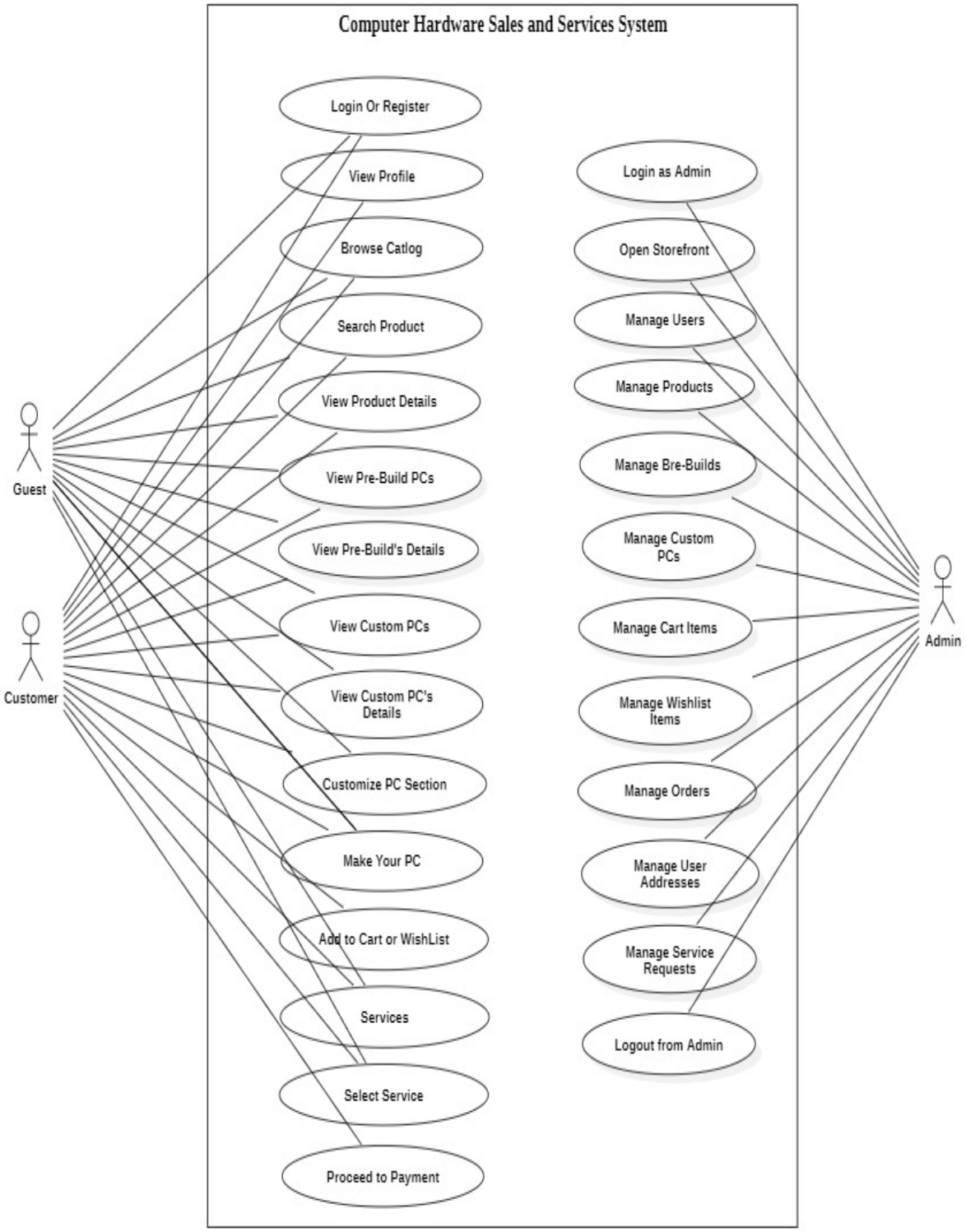


Figure 5.2 Use Case Diagram

5.3 USER AND ADMIN PROCESS FLOW DIAGRAM :

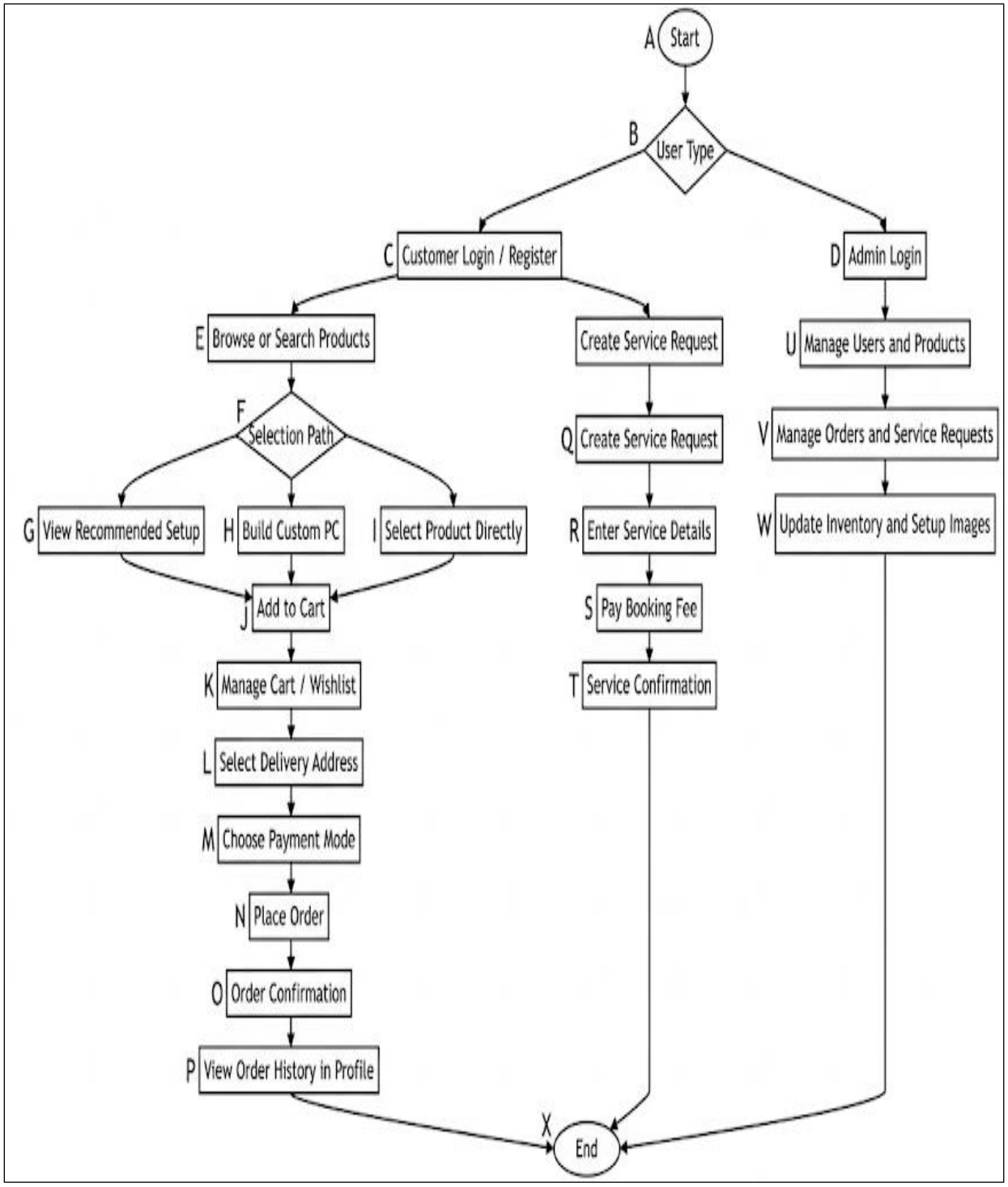


Fig 5.3 User and Admin Process Flow Diagram

In fig. 5.3 The diagram shows the workflow of a Computer Hardware Shop Management System. The process starts with the user selecting the user type as either customer or admin. If the user is a customer, they log in or register and then browse or search for products. The customer can view recommended setups, build a custom PC, or directly select a product. After selecting products, they add them to the cart, manage the cart or Wishlist, choose a delivery address and payment mode, and place the order.

The system then confirms the order and the customer can view the order history in their profile. Customers can also create a service request by entering service details and paying the booking fee, after which they receive service confirmation. If the user is an admin, they log in and manage users and products, handle orders and service requests, and update inventory and setup images. After completing the tasks, the process ends.

5.4 ER DIAGRAM :

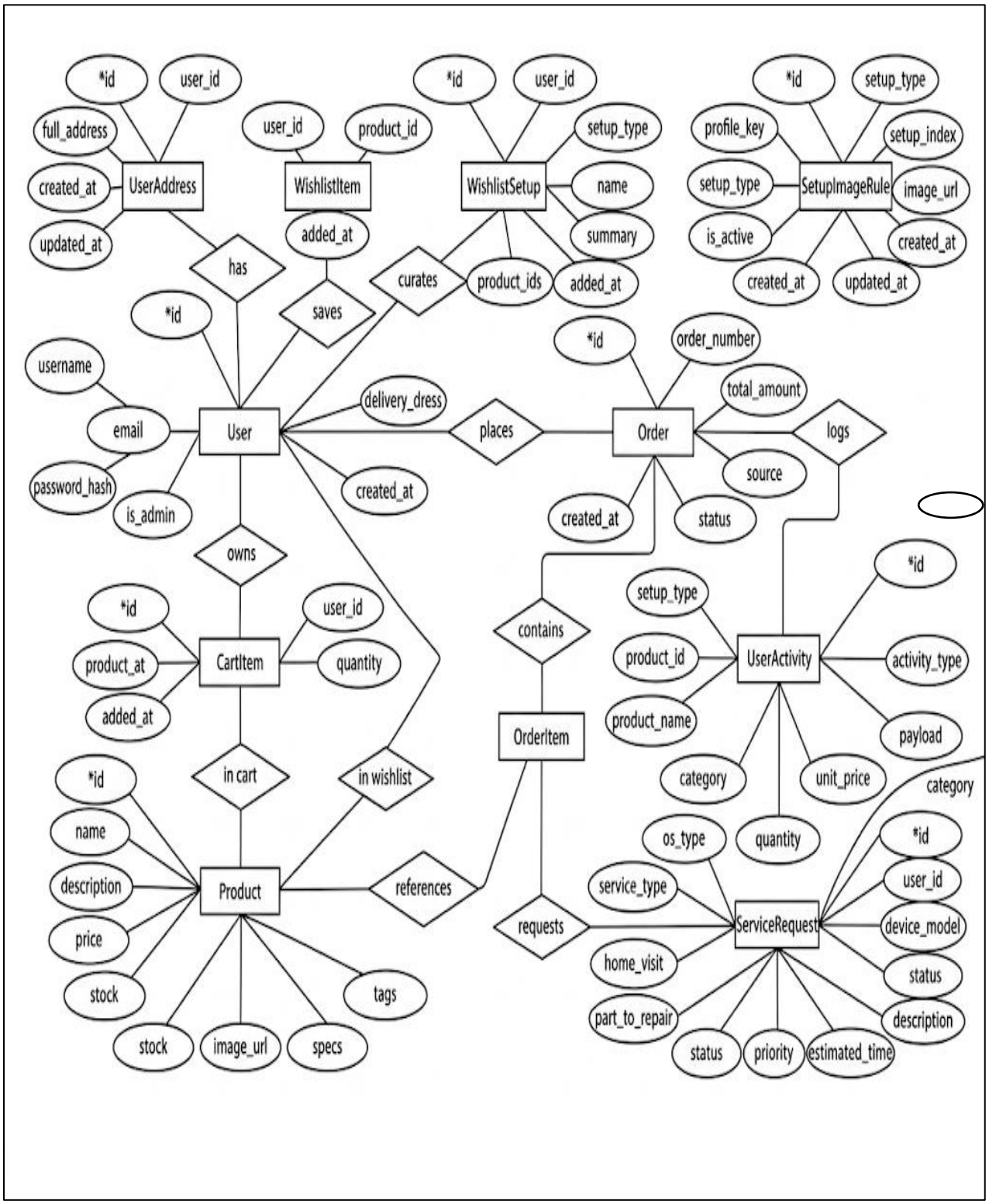


Fig. 5.4 ER Diagram

In fig 5.4 the diagram represents an entity relationship model of a Computer Hardware Shop Management System database. It shows different entities such as user, product, cart item, Wishlist item, Wishlist setup, order, order item, service request, user activity, and user address. The user entity stores information like username, email, password, and admin status and is connected to addresses, cart items, and Wishlist items. Products contain details such as name, description, price, stock, image, specifications, and tags, and they can appear in the cart or Wishlist. Users can place orders that contain multiple order items with product details, quantity, and price. The system also tracks service requests for repairs or home visits with details like status, priority, and estimated time.

5.5 DATABASE TABLE DESCRIPTION :

Figure 5.5 Database Table Description

Table Name	Primary Key	Key Columns (Datatype)	Purpose
user	id	username (VARCHAR80), email (VARCHAR120), password_hash (VARCHAR256), is_admin (BOOLEAN) created_at (DATETIME)	Stores login/account and admin role
user_address	id	user_id (INTEGER), full_address (VARCHAR500), created_at, updated_at	Stores saved customer addresses
product	id	name (VARCHAR200), category (VARCHAR50), price (FLOAT), stock (INTEGER), image_url, specs, tags	Product catalog and inventory
cart_item	id	user_id (INTEGER), product_id (INTEGER), quantity (INTEGER), added_at	Shopping cart items
wishlist_item	id	user_id (INTEGER), product_id (INTEGER), added_at	Wishlist items
customer_order	id	user_id, order_number (VARCHAR40), total_amount (FLOAT), payment_mode, delivery_address, status, created_at	Order header/master data
order_item	id	order_id (INTEGER), product_id (INTEGER), product_name, category, unit_price, quantity	Order line items
service_request	id	user_id, service_type, device_model, issue_description, status, priority, home_visit, address, created_at	Repair/installation/upgrade service workflow
user_activity	id	user_id, activity_type, summary, payload, created_at	User action logs
setup_image_rule	id	setup_type, profile_key, setup_index, setup_name, image_url, is_active, updated_at	Setup image configuration rules
alembic_version *(metadata)*	version_num	version_num (VARCHAR32)	Migration version tracking

In fig 5.5 The table shows the database structure of a Computer Hardware Shop Management System. It includes different tables such as user, user_address, product, cart_item, wishlist_item, customer_order, order_item, service_request, user_activity, and setup_image_rule. Each table has a primary key and important columns with specific data types to store information. The user table stores account details and admin roles, while the user_address table saves customer addresses. The product table maintains product catalog and inventory details. Cart_item and wishlist_item store products added to the cart and wishlist. Customer_order and order_item manage order information and purchased products. The service_request table records repair or installation requests from users. The user_activity table logs user actions in the system, and the setup_image_rule table manages setup image configurations.

- **Functionality provided by System :**

1. User Management
2. Product Management
3. Custom PC Setup Management
4. Cart and Order Management
5. Service Request Management
6. Inventory Management
7. Recommendation System
8. Report and Activity Tracking
9. Search and Product Browsing
10. Admin Management of Orders and Services

CHAPTER 06

RESULT AND APPLICATION

6.1 RESULT :

- **Login Page :**

The system developed was tested for our batch of final year diploma in computer engineering. students and the web pages and results are present year.

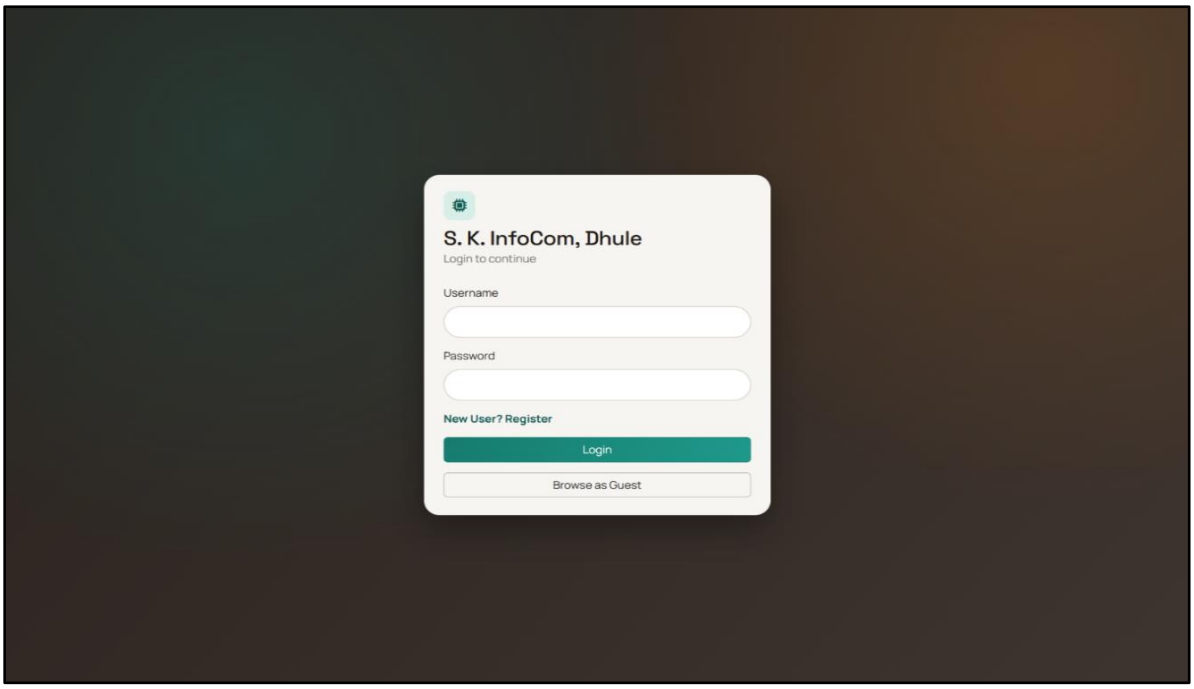


Figure 6.1 Login Page

In this figure 6.1, users are presented with a simple login form where they must enter a Username and Password to access their account. There are options for new users to Register, and existing users can simply click Login. Additionally, users have the choice to Browse as Guest without logging in. The interface is clean and functional, likely designed for quick access to the system.

- **Home Page :**

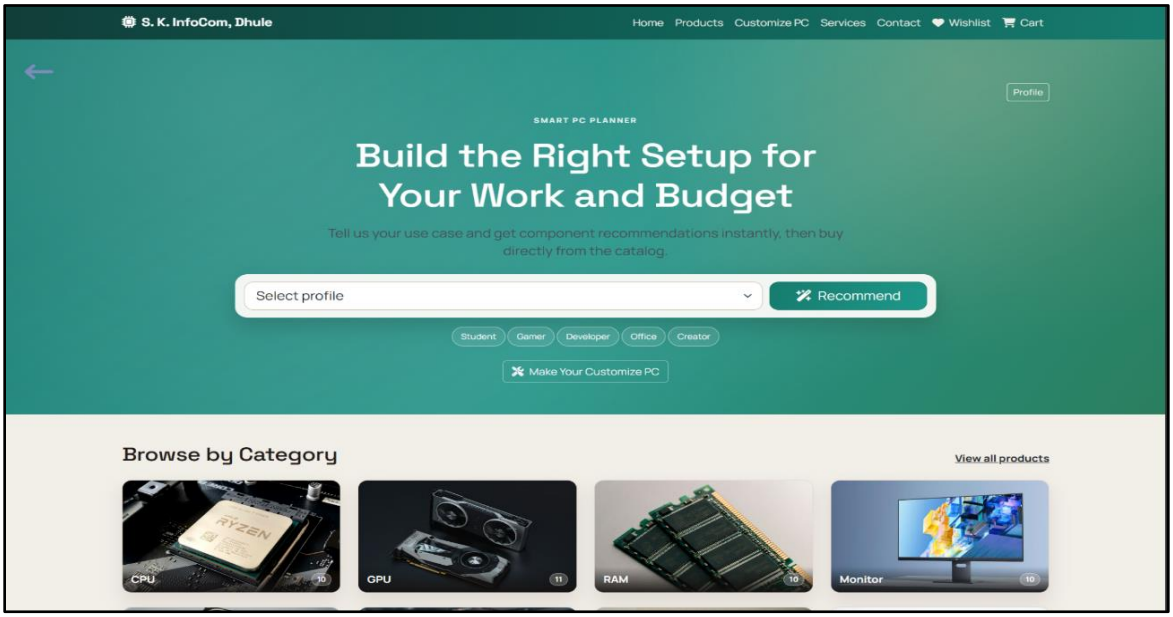


Figure 6.2 Home Page

In this figure 6.2, the login screen for S. K. InfoCom, Dhule is displayed, requiring users to enter a Username and Password to continue. It includes options for new users to Register, existing users to Login, and a link to browse the site as a Guest.

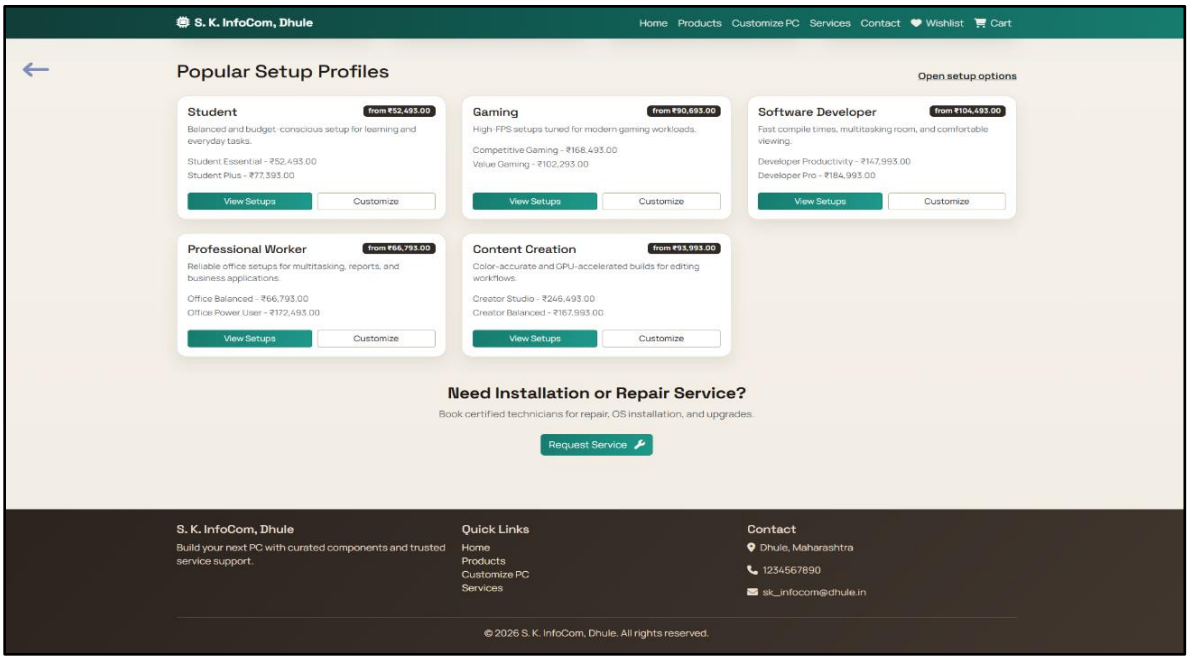
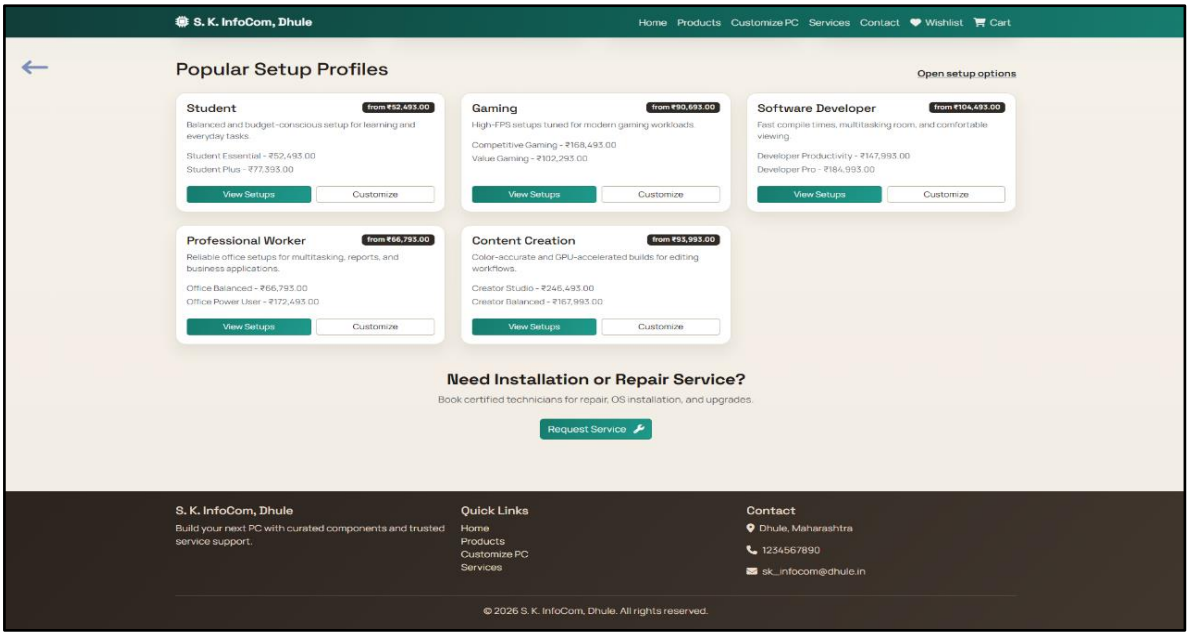


Figure 6.3 Popular Set-Ups and Contacts

In this figure 6.3, the S. K. InfoCom, Dhule homepage is shown, featuring a "Smart PC Planner" tool to help users build the right setup. The interface includes a navigation bar, profile access, popular setup profiles for different user types, and a footer with quick links and contact information.

- **Pre-Build PCs Page :**



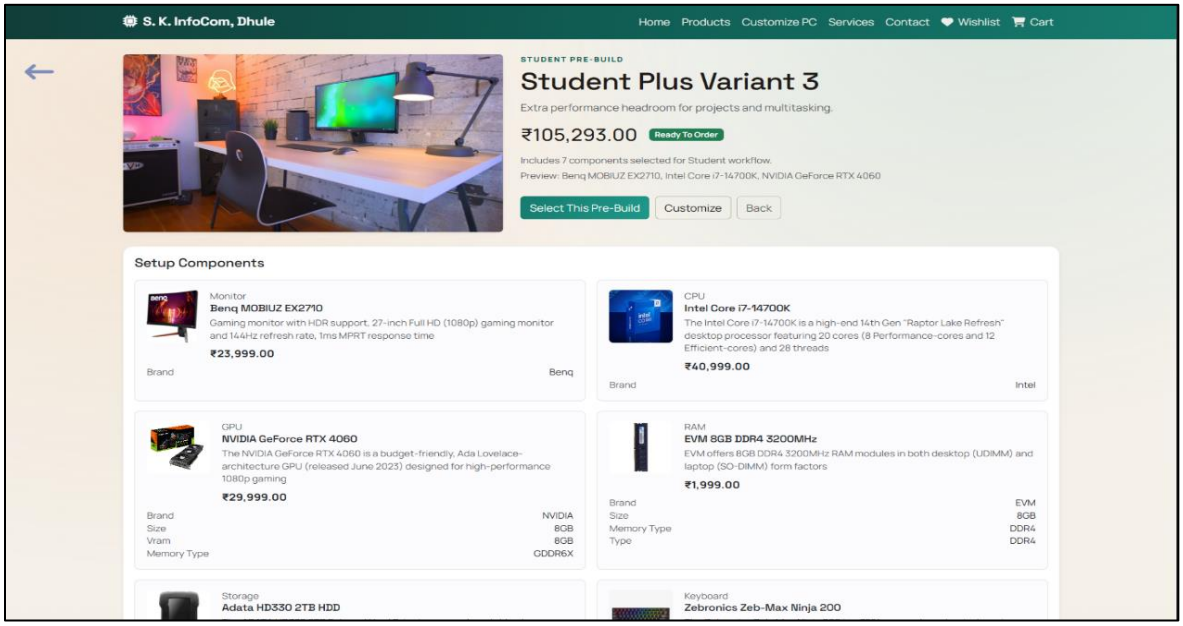


Figure 6.3 Pre-Build PCs with Detail Page

In this figure 6.4, the "Recommended Pre-Builds" section is presented, specifically highlighting Student setups. It showcases three different pre-built computer configurations, including the "Student Essential" and "Student Plus," listing their core components like the monitor, CPU, and GPU along with their prices.

In this figure, a detailed view of a specific pre-build, the "Student Plus Variant 3," is displayed. It breaks down the individual components included in the setup, such as the CPU, GPU, RAM, and storage, with their respective brands, specifications, and prices, culminating in a total price.

- **PC Customizing Page :**

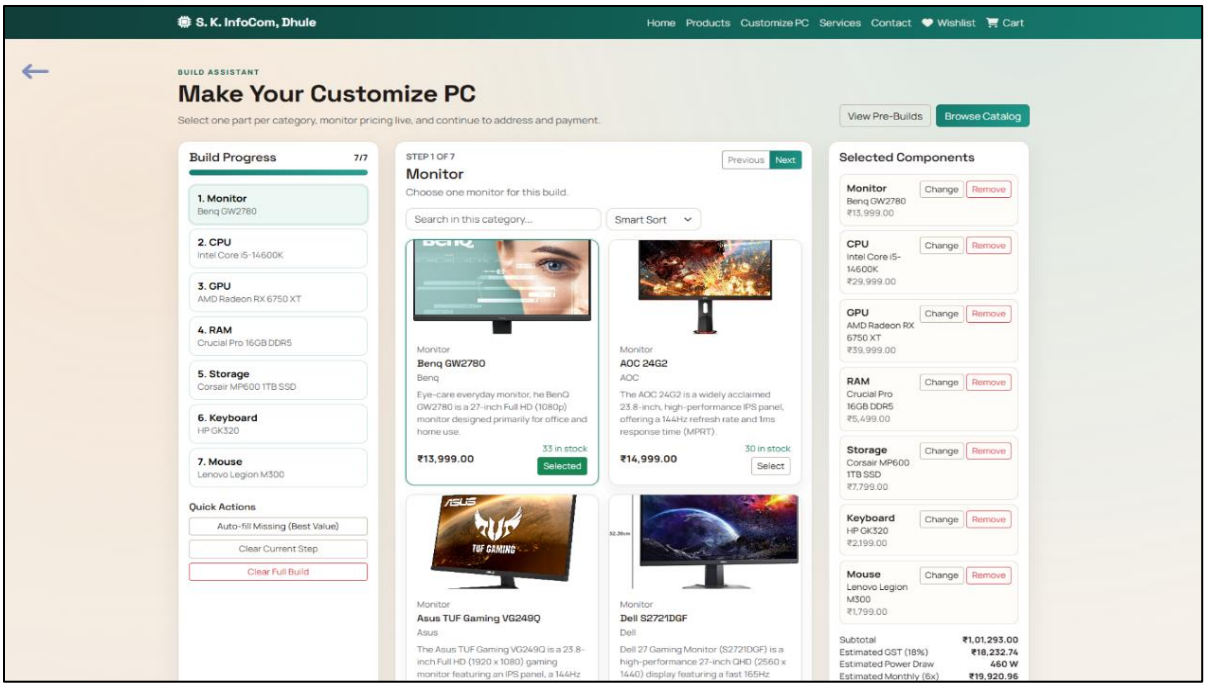
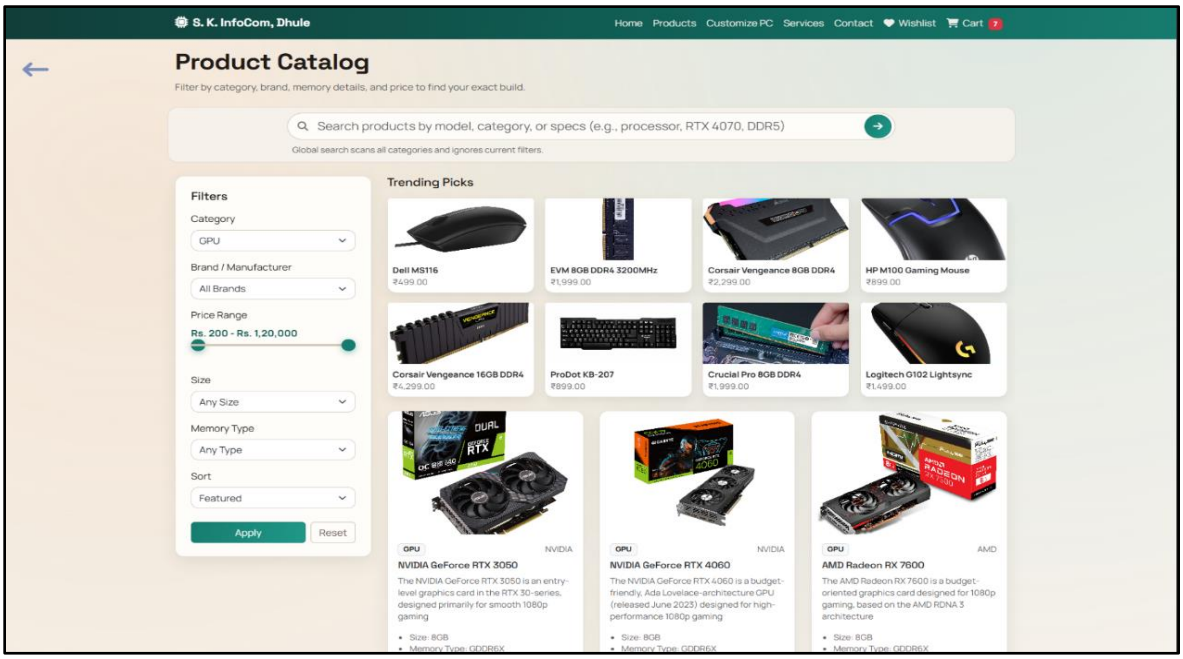


Figure 6.4 PC Customizing Page

In this figure 6.4, the "Make Your Customize PC" build assistant interface is shown. It allows users to manually select components for a custom computer from categories like Monitor, CPU, and GPU, with a live-updating list of selected components and their cumulative price on the right.

- **Product Catalog Page and Detail Page :**



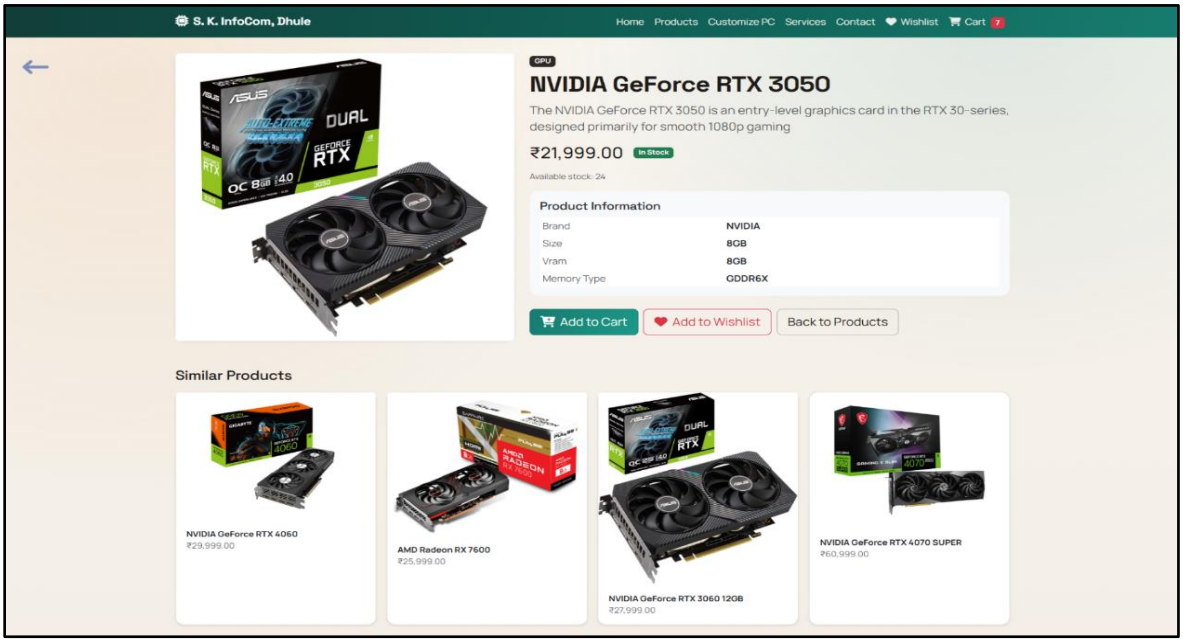
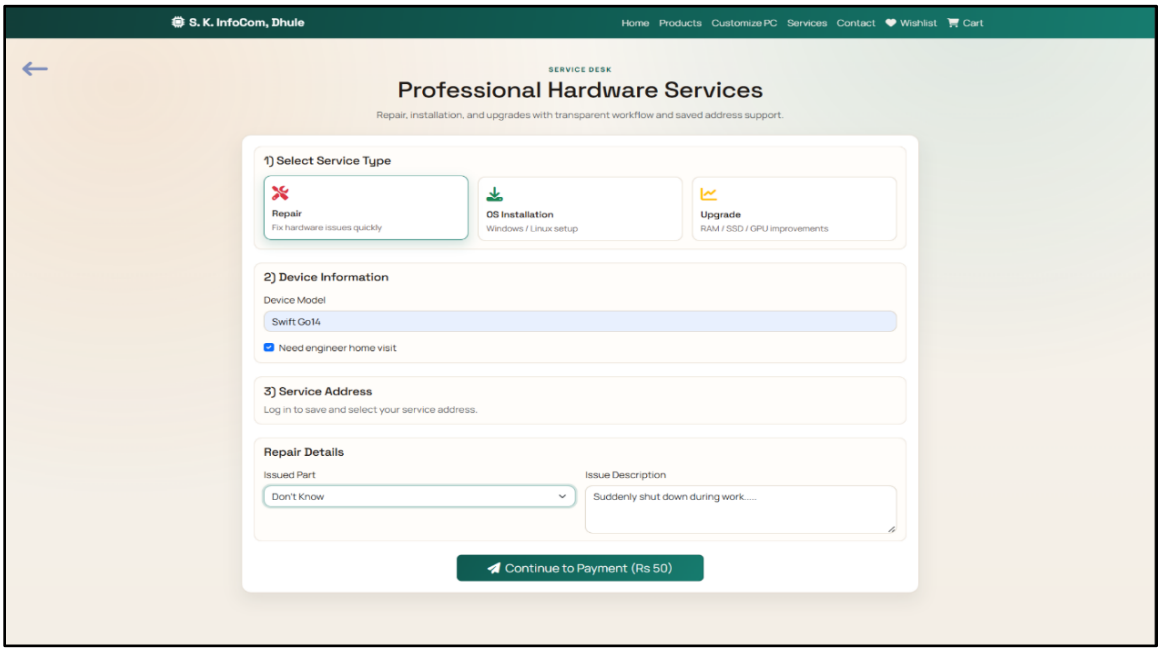


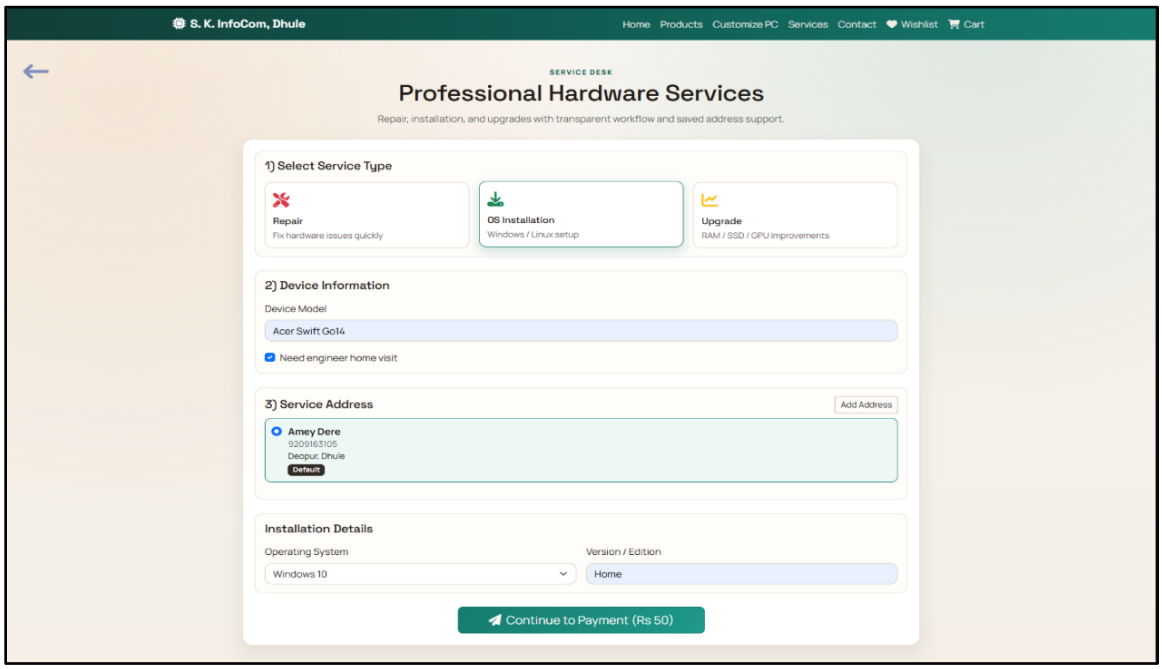
Figure 6.5 Product Catalog and detail Page

In this figure 6.5, the product catalog page is displayed, which includes filters for narrowing down products by category, brand, and price. It also shows a list of trending picks and detailed listings for specific products like the NVIDIA GeForce RTX graphics cards.

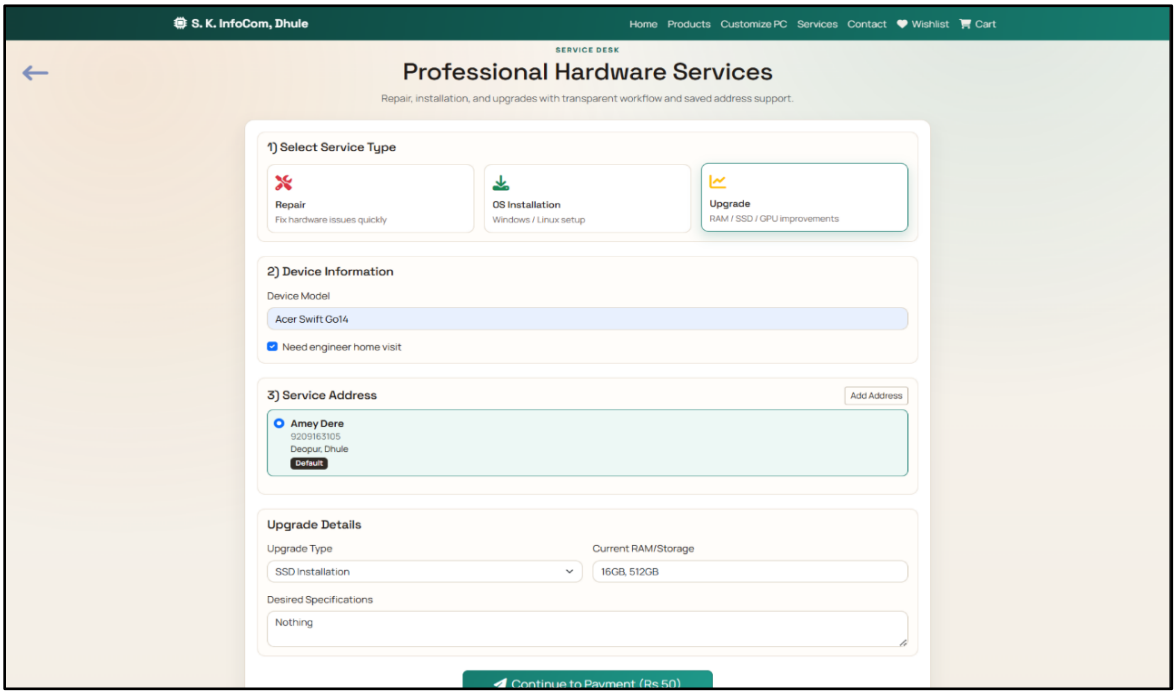
- **Repair and Service Page :**



In this figure 6.5, a professional hardware service form is displayed, allowing users to book services like Repair, OS Installation, or Upgrade. It includes steps for selecting service type, entering device details, and providing a service address, with an option to continue to payment.



In this figure 6.6, a professional hardware service form is displayed for requesting repairs, OS installation, or upgrades. It includes steps to select a service type, enter device details, choose a service address, and specify installation preferences like the operating system, with an option to continue to payment.



In this figure 6.7, a professional hardware service form is shown for booking repairs, OS installation, or upgrades. It includes sections for selecting service type, entering device details, choosing a service address, and specifying upgrade preferences, with an option to download related information.

- **Service Approval Page :**

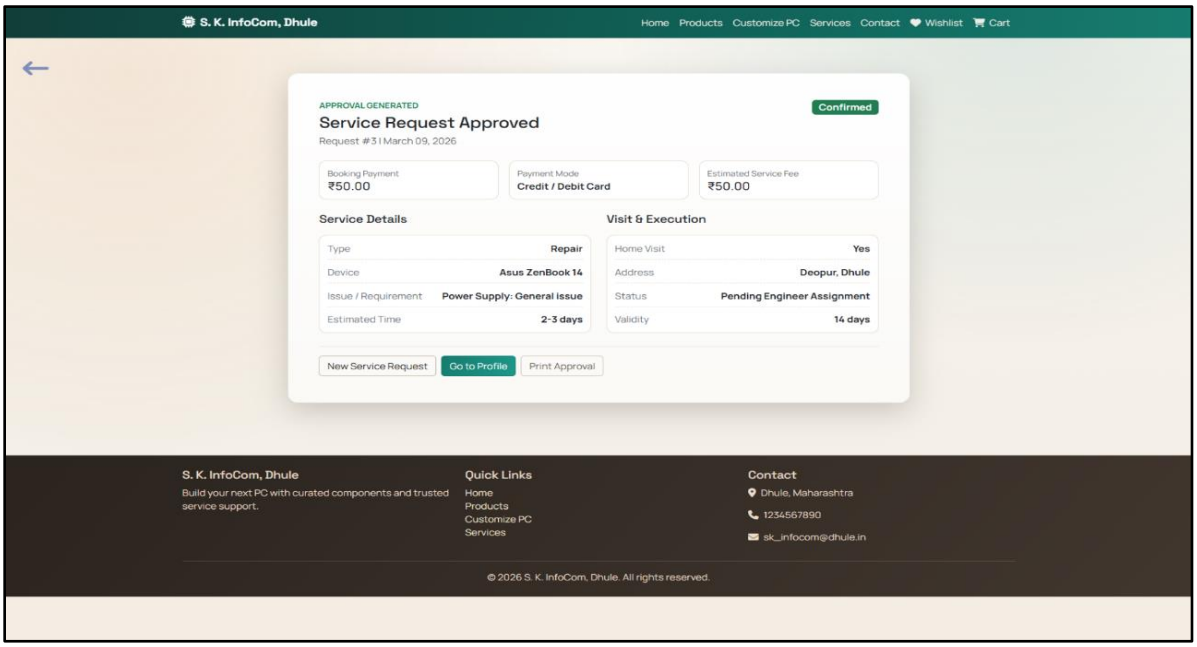
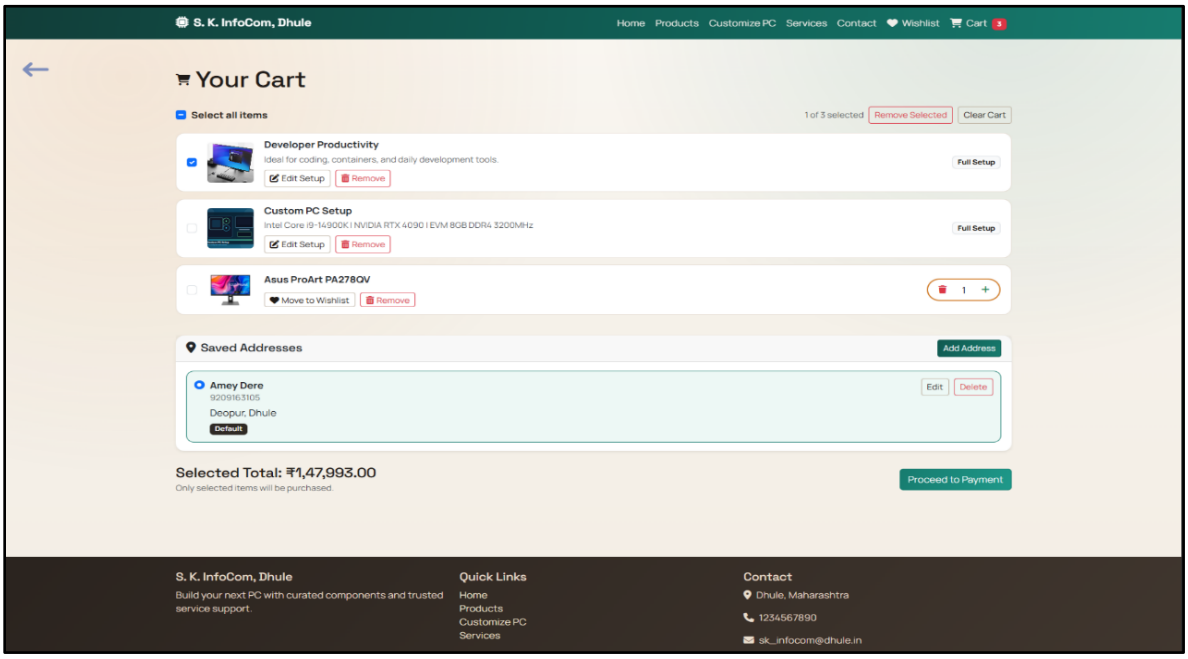


Figure 6.8 Product Catalog and detail Page

In this figure, a service request approval screen is shown, confirming a repair request with payment details, device information, visit status, and options to print or go to profile.

- **Cart Page :**



In this figure 6.9, the user's shopping cart is displayed, showing selected items like a Developer Productivity setup, a custom PC, and a monitor. It includes options to edit or remove items, a saved address section, and the selected total price for purchase. The cart allows users to select specific items for checkout rather than purchasing all at once. A default saved address is shown for delivery, with the total amount clearly displayed as ₹1,47,993.00. The interface also provides quick access to contact information and navigation options like Service, Warranty, and Cart.

- **Add Address page :**

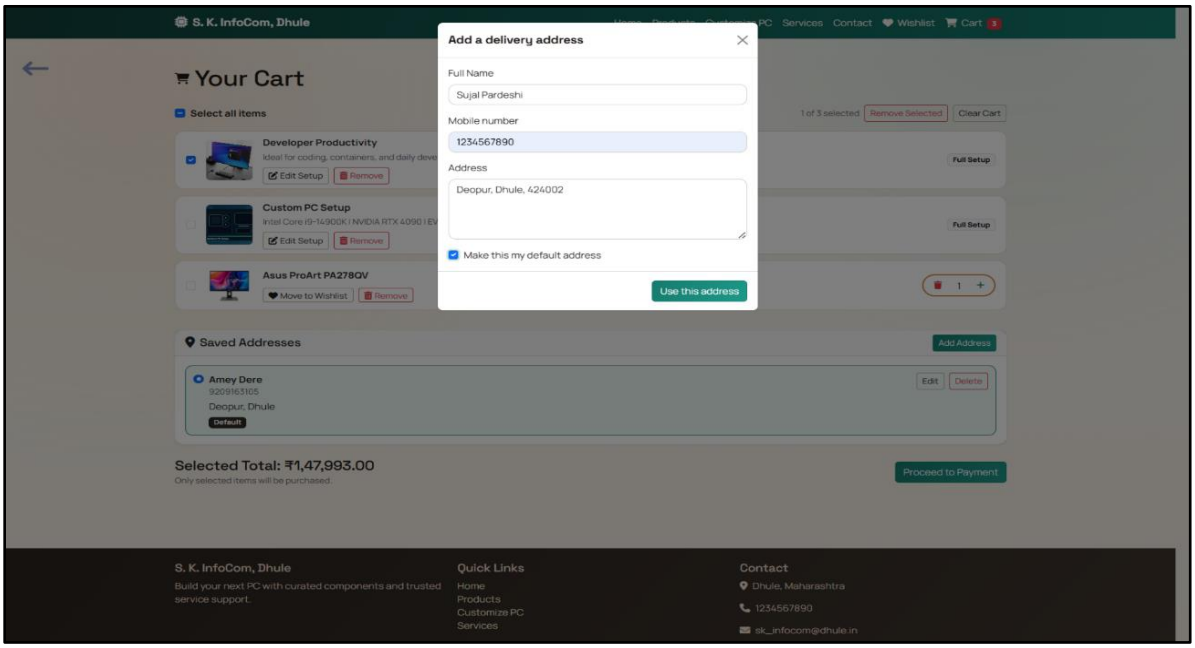
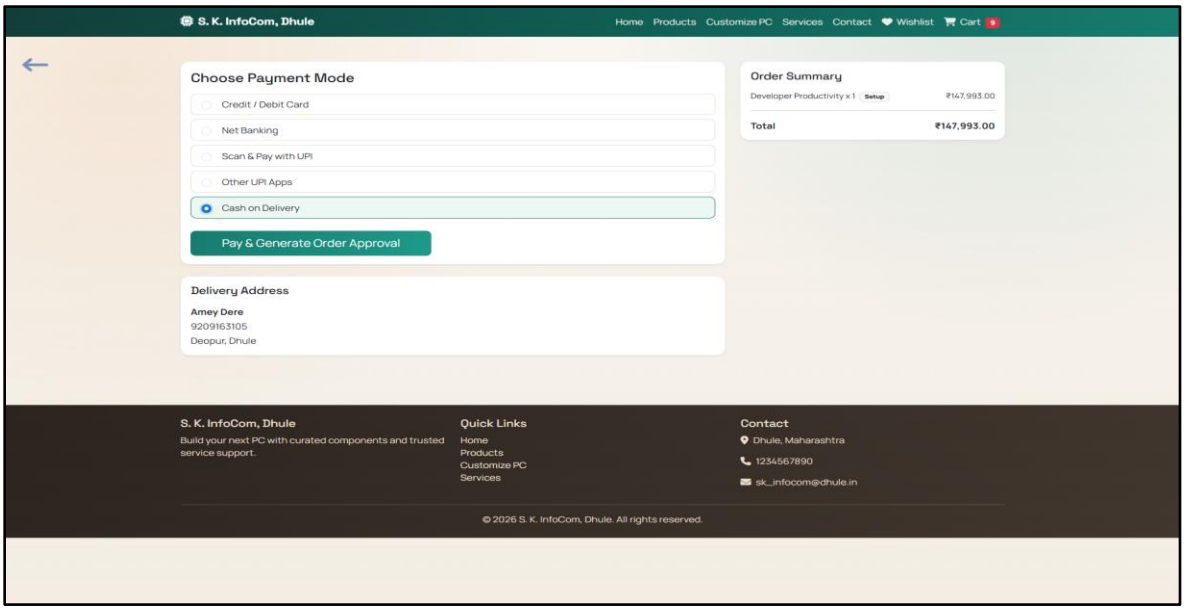


Figure 6.10 Product Catalog and detail Page

In this figure 6.10, the shopping cart page is displayed, showing items like Developer Productivity and Custom PC Setup with options to add to cart or edit. It includes a saved addresses section, a form to add a new delivery address, and the selected total amount for the order. Users can conveniently switch between multiple saved addresses or enter a new one using the provided form fields. The interface also displays contact information and service-related options like Warranty and Support for easy access. A clear total amount helps users review their costs before proceeding to checkout and payment.

- **Payment Range and Order Approval:**



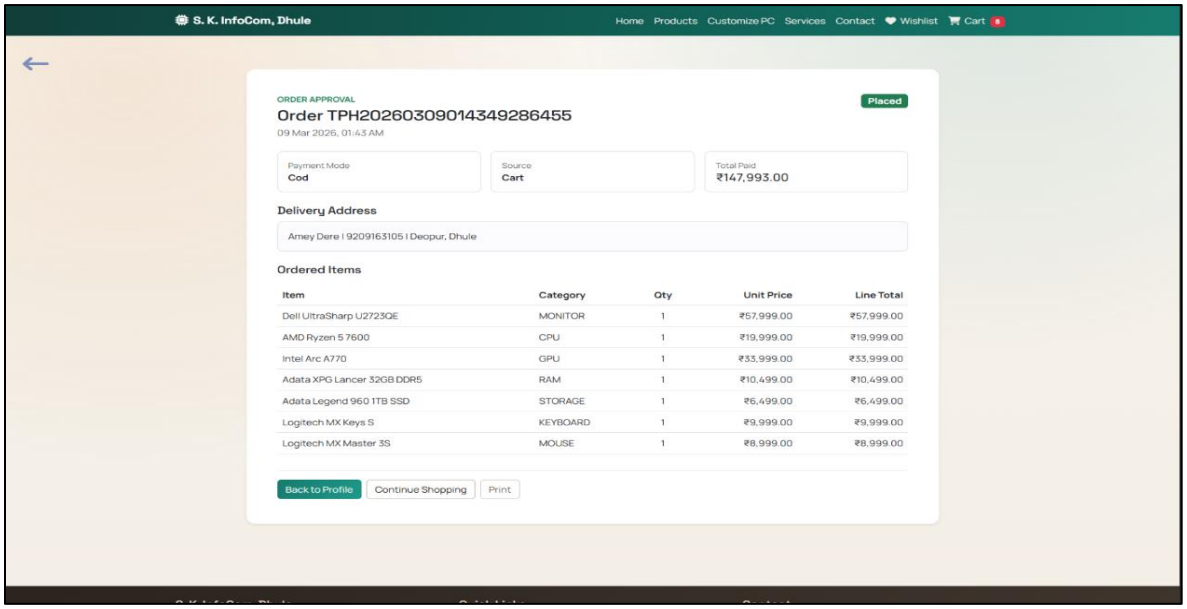
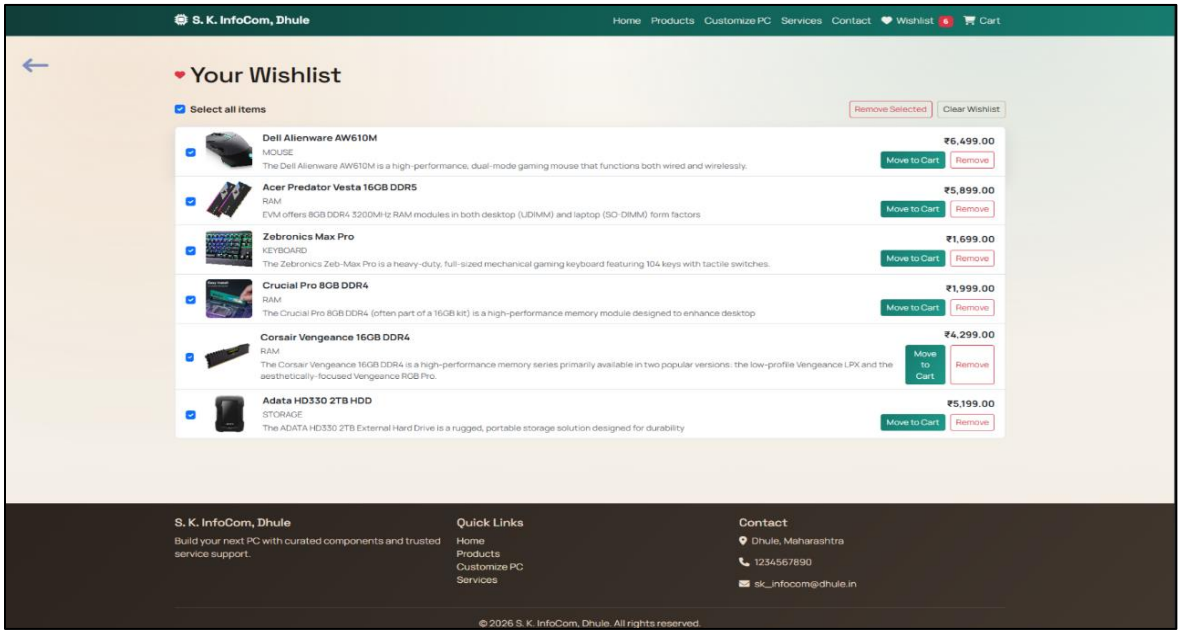


Figure 6.11 Payment Range and Order Approval Page

In these figures 6.11, the payment and order approval screens are shown, where users can select payment modes like card or UPI and view order summaries. The approval screen confirms the transaction with order details, itemized costs, and delivery address for the purchased setup.

- **Wishlist Page :**



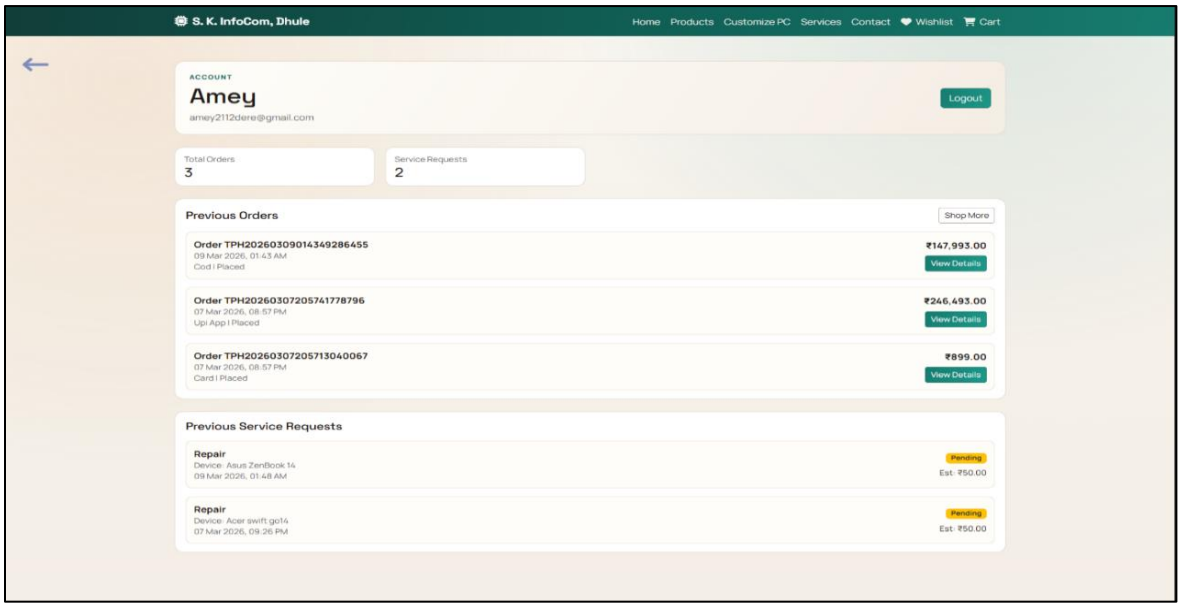


Figure 6.12 Wishlist and User Profile Page

Figure 6.12 Wishlist PageIn these figures, the user's wishlist and profile dashboard are displayed, showing saved items with prices and descriptions alongside order history and service request summaries. The profile page tracks total orders, service requests, and provides quick access to past transactions and logout functionality.

- **Admin Page :**

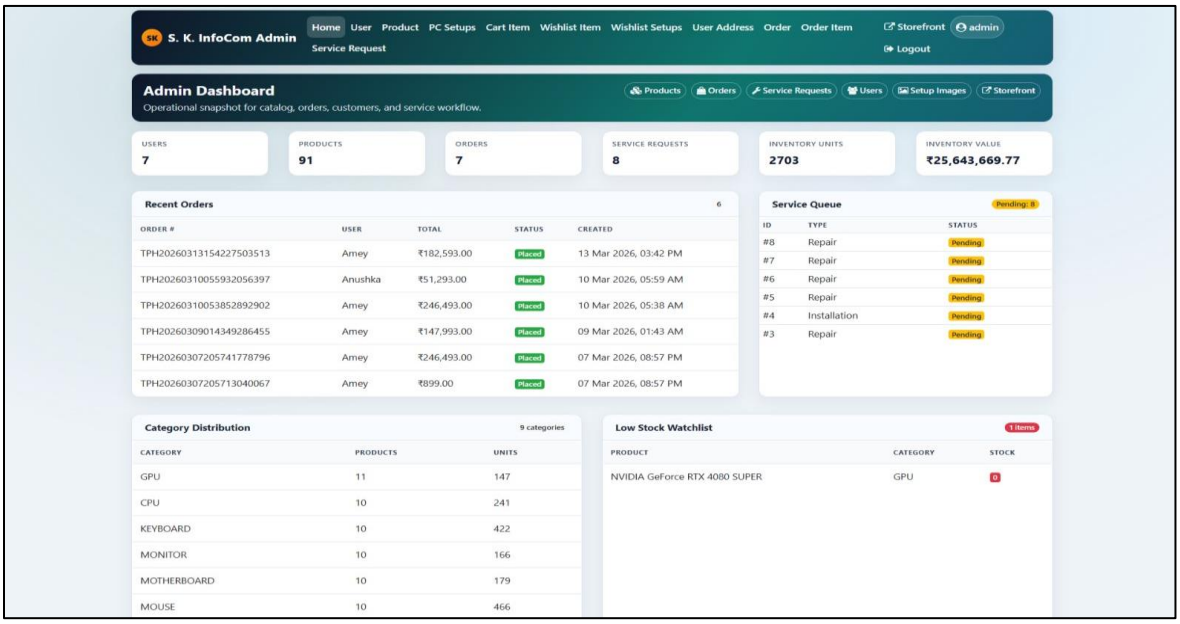
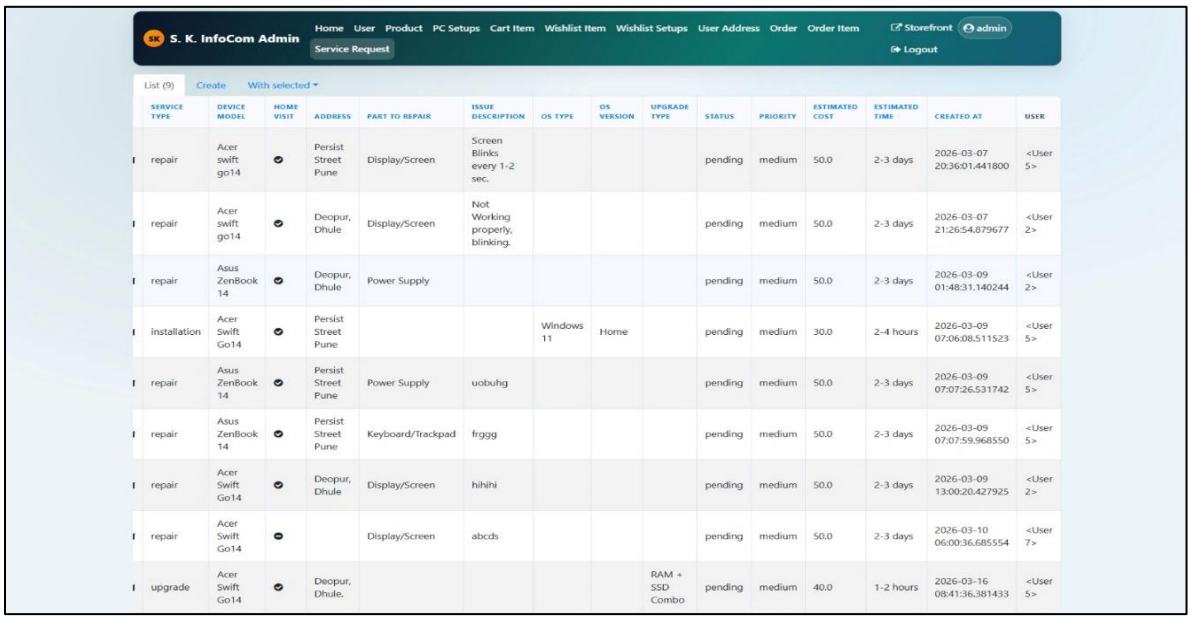


Figure 6.13 Admin Page

In this fig 6.13, admin dashboard shows a summary of users, products, orders, service requests, and inventory details. It also displays recent orders, service queue status, category distribution, and low stock items.

- **Service Page :**



SERVICE TYPE	DEVICE MODEL	HOME VISIT	ADDRESS	PART TO REPAIR	ISSUE DESCRIPTION	OS TYPE	OS VERSION	UPGRADE TYPE	STATUS	PRIORITY	ESTIMATED COST	ESTIMATED TIME	CREATED AT	USER
repair	Acer swift go14	✓	Persist Street Pune	Display/Screen	Screen Blinks every 1-2 sec.				pending	medium	50.0	2-3 days	2026-03-07 20:36:01.441800	<User 5>
repair	Acer swift go14	✓	Deopur, Dhule	Display/Screen	Not Working properly, blinking.				pending	medium	50.0	2-3 days	2026-03-07 21:26:54.879677	<User 2>
repair	Asus ZenBook 14	✓	Deopur, Dhule	Power Supply					pending	medium	50.0	2-3 days	2026-03-09 01:48:31.140244	<User 2>
installation	Acer Swift Go14	✓	Persist Street Pune			Windows 11	Home		pending	medium	30.0	2-4 hours	2026-03-09 07:06:08.511523	<User 5>
repair	Asus ZenBook 14	✓	Persist Street Pune	Power Supply	uobuhg				pending	medium	50.0	2-3 days	2026-03-09 07:07:26.531742	<User 5>
repair	Asus ZenBook 14	✓	Persist Street Pune	Keyboard/Trackpad	frggg				pending	medium	50.0	2-3 days	2026-03-09 07:07:59.968550	<User 5>
repair	Acer Swift Go14	✓	Deopur, Dhule	Display/Screen	hihihi				pending	medium	50.0	2-3 days	2026-03-09 13:00:20.427925	<User 2>
repair	Acer Swift Go14	✓		Display/Screen	abcds				pending	medium	50.0	2-3 days	2026-03-10 06:00:36.685554	<User 7>
upgrade	Acer Swift Go14	✓	Deopur, Dhule.					RAM + SSD Combo	pending	medium	40.0	1-2 hours	2026-03-16 08:41:36.381433	<User 5>

Figure 6.14 Service Page

CHAPTER 07

TESTING AND EVALUATION

6.1 TEST PLAN :

1. Document Information :

- Project: Web-Based Hardware MarketPlace For S. K. Infocom
- Application Type: Flask web application (server-rendered templates + API endpoints)
- Prepared On: 2026-03-30
- Version: 1.0
- Purpose:
  - Define a structured test strategy
  - Validate functional correctness
  - Ensure integration stability
  - Confirm release readiness

2. Test Objectives :

- Verify complete user flow (login → purchase → order completion)
- Validate all modules:
  - Cart
  - Wishlist
  - Orders
  - Recommendations
  - Custom PC builder
  - Service requests
- Ensure data integrity using SQLAlchemy models
- Detect bugs early using:
  - Smoke testing
  - Regression testing

3. System Under Test (SUT) :

3.1 Core Technology Stack :

- Python Flask:
  - Flask
  - Flask-SQLAlchemy
  - Flask-Login
  - Flask-Migrate
  - Flask-Admin
- Frontend:
  - HTML templates (templates/)
  - CSS & JS (static/)
- Database:
  - SQLAlchemy models (models.py)

3.2 Major Functional Modules :

- Authentication:
  - /register, /login, /logout, /profile
- Product Browsing:
  - /products, /product/<id>, /search
- Cart:
  - /cart, /api/cart/*, /cart/setup/<group_id>
- Wishlist:
  - /wishlist, /api/wishlist/*
- Checkout & Payment:
  - /proceed-payment, /payment
- Orders:
  - Order history routes
- Custom PC Builder:
  - /custom-pc, /custom-pc/checkout
- Service Requests:
  - /service, /service/payment

4. Scope :

4.1 In Scope :

- Functional testing of:
  - All pages
  - API endpoints
- Integration testing (routes + database)
- Form validation and negative testing
- Cross-module workflows:
  - Product → Cart → Payment → Order
- Regression testing

4.2 Out of Scope :

- Performance/load testing
- External penetration testing
- Network/infrastructure testing
- Mobile application testing

5. Test Strategy :

5.1 Test Levels :

- Unit Testing
  - Utility functions
  - Validation logic
- Integration Testing
  - Routes + Database
- End-to-End Testing
  - Full user journeys
- Manual Testing
  - UI and usability

5.2 Test Types :

- Functional testing
- Negative testing
- Regression testing
- Basic security testing
- Browser compatibility testing

5.3 Tools :

- pytest (unit + integration testing)
- pytest-cov (coverage)
- Playwright / Selenium (E2E testing)
- Manual checklist testing

6. Test Environment :

6.1 Environments

- Local QA environment
- Staging environment

6.2 Configuration

- Python environment (requirements.txt)
- Database setup with migrations
- Seeded product data

6.3 Supported Browsers

- Google Chrome
- Microsoft Edge
- Mozilla Firefox

7. Test Data Plan :

7.1 User Data

- Normal user
- Admin user
- Invalid login credentials

7.2 Product Data

- In-stock products
- Out-of-stock products
- Products with detailed specifications

7.3 Transaction Data

- Single-item cart
- Multi-item cart
- Wishlist data
- Multiple addresses
- Completed orders

8. Entry and Exit Criteria :

8.1 Entry Criteria :

- Application runs successfully
- Database is updated
- Test data is available
- No major environment issues

8.2 Exit Criteria :

- All **P0 & P1 test cases pass**
- No critical bugs remain
- All key flows work correctly
- Regression success rate ≥ 95%

9. Defect Priority Levels :

- P0 (Critical):
  - Login failure
  - Payment failure
  - System crash
- P1 (High):
  - Major feature not working
- P2 (Medium):
  - Minor functionality issues
- P3 (Low):
  - UI or cosmetic issues

10. Test Scenario Categories :

- Authentication scenarios
- Profile management
- Product browsing
- Search functionality
- Cart operations
- Wishlist operations
- Address management
- Checkout & payment
- Orders & history
- Recommendations
- Prebuilt systems
- Custom PC builder
- Service requests
- Security testing
- Admin access
- Image handling

11. Regression Testing :

11.1 Smoke Testing

- Login
- Product listing
- Add to cart
- Checkout
- Order creation

12. Non-Functional Testing :

- Page load performance
- UI responsiveness
- Session stability
- Error message clarity

13. Risks & Mitigation :

- Risk: Missing test data
  - Solution: Maintain seed data scripts
- Risk: Complex module integration
  - Solution: Dedicated integration tests
- Risk: Large codebase changes
  - Solution**:** Run smoke tests frequently

14. Deliverables :

- Test plan document
- Test cases
- Bug reports
- Test summary report

15. Approval Checklist

- Test coverage verified
- All critical flows tested

6.2 LOGIN FORM TEST CASES :

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_01	Valid login	Correct username & password	Login successful	Login successful	Pass
TC_02	Wrong password	Correct username & wrong password	Error message should be shown	Error message shown	Pass
TC_03	Wrong username	Wrong username & correct password	Error message should be shown	Error message shown	Pass
TC_04	Empty fields	Leave username & password blank	Validation message should be shown	Validation message shown	Pass
TC_05	Empty password	Enter username only	Password required message	Password required message	Pass
TC_06	Empty username	Enter password only	Username required message	Username required message	Pass
TC_08	Case sensitivity check	Username in wrong case & correct password	System should be case-sensitive	Login failed	Pass
TC_09	Password masking	Enter password	Password should be masked	Password hidden as dots	Pass
TC_10	Remember me option	Select "Remember Me" & login	Session should be saved	User remains logged in	Pass

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_11	Session timeout	Stay inactive after login	Session should expire after timeout	Auto logout	Pass
TC_12	Login button functionality	Click login button	Button should work correctly	Action performed	Pass
TC_13	Enter key login	Press Enter after input	Should login using Enter key	Login triggered	Pass
TC_14	Multiple failed attempts	Enter wrong password multiple times	System should restrict access	Still allows login attempts	Fail

- **<u>Defect Report 1</u> :**

Field	Details
Defect ID	DR_LGN_002
Project Name	Web Bases Hardware Marketplace for S. K. Infocom
Application Type	Web Application
Release Version	v1.0
Module	Login Module
Detected Build Version	v1.1
Summary	Unlimited login attempts allowed

Field	Details
Description	During testing, it was observed that the system does not restrict multiple failed login attempts. Users can continuously attempt login without any limitation, which may lead to security risks such as brute-force attacks.
Actual Result	Unlimited login attempts allowed
Expected Result	System should restrict attempts or show warning/lock account after multiple failures
Severity	High
Priority	High
Status	Open
Reported By	Tester
Reported Date	08/04/2026

6.3 SEARCHING FUNCTIONALITY :

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_01	Verify page loads successfully	Open page URL	Page should load without errors	Page loads	Pass
TC_02	Verify all setup cards are displayed	No input	Student, Gaming, Developer, etc. display	All setup cards visible	Pass

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_03	Verify footer links	Click Quick Links	Correct pages should open	Redirect works	Pass
TC_04	Verify "Customize" button	Click "Customize"	Custom PC builder should open	Redirects to customize page	Pass
TC_05	Verify pricing display	View price section	Correct price format displayed	Prices visible	Pass
TC_06	Verify setup description	Read card description	Proper description displayed	Description shown	Pass
TC_07	Verify navigation bar links	Click Home/Products/ Services	Correct pages should open	Redirect works	Pass
TC_08	Verify Wishlist button	Click Wishlist	Wishlist page should load	Page opens	Pass
TC_09	Verify Cart button	Click Cart	Cart page should load	Page opens	Pass
TC_10	Verify "Request Service" button	Click button	Service page should open	Redirects	Pass
TC_11	Verify responsiveness	Resize browser	Page should be responsive	Layout adjusts	Pass
TC_12	Verify "View Setups" button	Click "View Setups" on any card	Correct setups page should open	Redirects to wrong page	Fail

- **Defect Report 2 :**

Field	Details
Defect ID	DR_UI_001
Project Name	Web Based Hardware Marketplace for S.K Infocom
Application Type	Web Application
Release Version	v1.0
Module	Navigation Module
Detected Build Version	v1.1
Summary	"View Setups" button redirects to incorrect page
Description	During testing, it was observed that when the user clicks on the "View Setups" button on any setup card, the system redirects to an incorrect or unrelated page instead of the intended setups page. This affects user navigation and overall user experience.
Actual Result	Redirects to wrong page
Expected Result	System should redirect to the correct setups page
Severity	Medium
Priority	High
Status	Open
Reported By	Tester
Reported Date	08/04/2026

6.4 SERVICE FUNCTIONALITY TEST CASES :

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_01	Verify page loads correctly	Open service page URL	Page should load without errors	Page loads	Pass
TC_02	Verify service type selection	Select "Repair"	Selected service type should be highlighted	Option selected	Pass
TC_03	Verify multiple service types	Select Installation/Upgrade	Only one service type at a time	Option changes	Pass
TC_04	Verify device model input	Enter "Swift Go14"	Input should accept valid device name	Text entered	Pass
TC_05	Verify empty device model	Leave blank	Validation message should appear	Error/continue	Pass
TC_06	Verify home visit checkbox	Check checkbox	Checkbox should toggle correctly	Checkbox selected	Pass
TC_07	Verify address section access	Not logged in	User should be asked to login	Prompt shown	Pass

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_08	Verify issue part dropdown	Select "Don't Know"	Dropdown should work correctly	Option selected	Pass
TC_09	Verify issue description input	Enter text	Input should accept description	Text displayed	Pass
TC_10	Verify empty description	Leave blank	Validation should be triggered	Error/continue	Pass
TC_11	Verify multiple clicks on payment	Click repeatedly	Should prevent duplicate action	Works once	Pass
TC_12	Verify button without required fields	Click without input	Should block and show validation	Error shown	Pass
TC_13	Verify navigation bar links	Click menu options	Correct pages should open	Redirect works	Pass
TC_14	Verify form reset behavior	Refresh page	All fields should reset	Form clears	Pass
TC_15	Verify UI responsiveness	Resize screen	Page should be responsive	Layout adjusts	Pass
TC_16	Verify special characters in input	Enter symbols	Should handle safely	Accepted/validated	Pass

Test Case ID	Objective	Input	Expected Output	Actual Output	Status
TC_17	Verify long text input	Enter large description	Should not break UI	Accepted	Pass
TC_18	Verify invalid navigation	Use back button	Should return safely	Page loads	Pass
TC_19	Verify payment amount display	View button	Correct amount should display	₹50 shown	Pass
TC_11	Verify multiple clicks on payment	Click repeatedly	Should prevent duplicate action	Works once	Pass
TC_20	Verify continue to payment button	Click button	Should navigate to payment page	Page does not redirect	Fail

- **Defect Report 3 :**

Field	Details
Defect ID	DR_SRV_001
Project Name	Web Based Hardware Marketplace for S.K Infocom
Application Type	Web Application
Release Version	v1.0

Field	Details
Module	Service Request / Payment Module
Detected Build Version	v1.1
Summary	"Continue to Payment" button not redirecting to payment page
Description	During testing, it was observed that when the user clicks on the "Continue to Payment" button after filling the service request form, the system does not redirect to the payment page. This prevents users from completing the service booking process.
Actual Result	Page does not redirect after clicking button
Expected Result	System should navigate to the payment page
Severity	High
Priority	High
Status	Open
Reported By	Tester
Reported Date	08/04/2026

CHAPTER 08

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION :

The web-based hardware marketplace for S.K. Infocom provides an efficient platform to sell and manage computer hardware products online. It allows customers to easily browse products, check details, and place orders from anywhere. The system improves inventory control, reduces manual work, and increases business reach. It also helps in maintaining accurate product records and managing orders efficiently. The platform saves time for both customers and the business. Overall, it supports the growth of S.K. Infocom by adopting modern digital technology.

8.2 FUTURE SCOPE :

1. The system can be developed into a mobile application for easier access on smartphones.
2. Online payment gateways can be added for faster and secure transactions.
3. The marketplace can support multiple vendors and suppliers to expand product variety.
4. Advanced data analytics and sales reports can be included to improve business decisions.
5. Integration with delivery and tracking systems can improve customer service and order management.

CHAPTER 09

REFERENCES AND BIBLIOGRAPHY

1. REFERENCES :

Websites :

- Gigisoft - https://digisoft.in
- MD Computers - https://mdcomputers.in
- PrimeABGB - https://www.primeabgb.com

Books :

- Title: The Online Marketplace: How Platforms Are Reshaping Retail
  Author: Daniel S. Evans.
- Title: Digital Business and E-Commerce Management
  Author: Dave Chaffey.

2. BIBLIOGRAPHY:

The project "Web Based Hardware Marketplace for S.K Infocom" is developed using technologies such as Flask, Python, HTML5, CSS3, JavaScript, and SQLite. Knowledge and implementation support were taken from official documentation and online learning resources. Basic software engineering concepts were referred from Ian Sommerville and Roger S. Pressman. Additional help was taken from various web development tutorials and practical references for building and testing the application.

Evaluation Sheet For The Micro Project

Academic Year : 2025 – 2026	**Name of Faculty :** Mr. A.V. Agrawal
Sem : Sixth	**Program Name And Code :** CO6K
Course Code : 316004	**Course name :** Capstone

Title of The Project : Web-Based Hardware Marketplace for S. K. Infocom

1.0 Introduction :

- **Cos addressed by the Micro Project :**
  CO1: Identify and explain the project problem.
  CO2: Analyze project feasibility and viability.
  CO3: Apply knowledge to solve real-world problems.
  CO4: Present the project results and findings
- **Major Learning Outcomes achieved by students by doing the Project :**
  a) **Practical Outcomes :**
     a. Develop a web-based platform for SK Infocom to sell and manage computer hardware products online.
     b. Improve product management, inventory tracking, and customer purchasing experience.
  b) **Unit Outcomes (in Cognitive domain) :**
     a. Understand the design and development of a web-based marketplace system.
     b. Apply web technologies to manage hardware products, users, and online transactions efficiently.
  c) **Outcomes in Affective Domain:**
     a. Develop teamwork and responsibility.
     b. Show interest in learning new web technologies.

- **Marks: Name of Student:- Amey Pravin Dere**

(A) Marks for Groupwork:	(B) Marks for Individual work:	(C) Total Marks (A + B) =

Mr. A. V. Agrawal
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    b. Apply web technologies to manage hardware products, users, and online transactions efficiently.
  c) Outcomes in Affective Domain:
    a. Develop teamwork and responsibility.
    b. Show interest in learning new web technologies.

- **Marks: Name of Student:- Shrutika Sandip Dhondge**

(A) Marks for Groupwork:	(B) Marks for Individual work:	(C) Total Marks (A + B) =

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    a. Develop teamwork and responsibility.
    b. Show interest in learning new web technologies.

- **Marks: Name of Student:- Anushka Sangram Digole**

(A) Marks for Groupwork:	(B) Marks for Individual work:	(C) Total Marks (A + B) =

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1.0 Introduction :

- **Cos addressed by the Micro Project :**
  CO1: Identify and explain the project problem.
  CO2: Analyze project feasibility and viability.
  CO3: Apply knowledge to solve real-world problems.
  CO4: Present the project results and findings
- **Major Learning Outcomes achieved by students by doing the Project :**
  a) Practical Outcomes :
    a. Develop a web-based platform for SK Infocom to sell and manage computer hardware products online.
    b. Improve product management, inventory tracking, and customer purchasing experience.
  b) Unit Outcomes (in Cognitive domain) :
    a. Understand the design and development of a web-based marketplace system.
    b. Apply web technologies to manage hardware products, users, and online transactions efficiently.
  c) Outcomes in Affective Domain:
    a. Develop teamwork and responsibility.
    b. Show interest in learning new web technologies.

- **Marks: Name of Student:- Sujal Nitin Pardeshi**

(A) Marks for Groupwork:	(B) Marks for Individual work:	(C) Total Marks (A + B) =

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